THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

December 12, 2022 TIME: 9:30 AM – 12:30 PM

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:
Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg·°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

IV. Ideal gas law constant = 0.082056 L·atm/(K·mol)
Partial pressure of CO_2 in atmosphere: $1 \times 10^{-3.5}$ atm
Henry's Law constant for CO_2: 0.033363 mol/L-atm
Acidity constants for H_2CO_3: $K_1 = 4.47\times10^{-7}$; $K_2 = 4.68\times10^{-11}$
Henry's Law constant for O_2 (25 °C) = 0.001263 mol/(L·atm)
Dissociation constant of water (25 °C) $K_w = 1\times10^{-14}$

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (10 multiple choice questions with 3 marks each)

1. Suppose the exhaust gas from an automobile contains 1% by volume of CO. What is this concentration in mg/m^3 at 25 °C and 1 atm?

   (a) 10,000 mg/m^3; (b) 11,445 mg/m^3; (c) 12,492 mg/m^3; or (d) 13,989 mg/m^3.

2. A heat engine receives a heat from a hot reservoir with a temperature of 600 °C for doing work. The waste heat is discharged into a cold reservoir with a temperature of 20 °C. What is the Carnot efficiency of this engine?

   (a) 97%; (b) 80%; (c) 66%; or (d) 60%.

3. What is the molar concentration of 10 g glucose ($C_6H_{12}O_6$) dissolved in 1 L of water?

   (a) 10 g/L; (b) 0.05 mmol/L; (c) 0.0555 M; or (d) 1 mol/L.

4. What is the **RIGHT** comparison for theoretical oxygen demand (ThOD), BOD, and COD for a certain organic pollutant?

   (a) COD>ThOD>BOD; (b) ThOD>BOD>COD; (c) ThOD>COD>BOD; or (d) BOD>COD>ThOD.

5. Which of the following pollutant is **NOT** a chemical pollutant?

   (a) lead; (b) disinfection byproduct; (c) nitrogen; or (d) pathogen.

6. A sample of groundwater has 150 mg/L of Ca^{2+} and 60 mg/L of Mg^{2+}. What is the total hardness expressed in mg/L as $CaCO_3$?

   (a) 621.0; (b)12.4; (c) 550.0; or (d) 15.2.

7. A circular primary sedimentation tank receives a wastewater influent of 24,000 m^3/d. The designed surface overflow rate for the tank is 2.5 m/h while the HRT is 2 hours. What is the designed depth of the tank?

   (a) 2.0 m; (b) 3.0 m; (c) 4.0 m; or (d) 5.0 m.

8. Which is **NOT** the key factor affecting the food over microorganism ratio in activated sludge process?

   (a) influent TSS; (b) influent BOD; (c) influent flow rate; or (d) biomass concentration.

9. Which is **NOT** a criteria air pollutant?

   (a) lead; (b) CO_2; (c) NO_2; or (d) ground-level ozone.

10. Which of the following statements is **NOT** true?

    (a) ocean has a slightly alkaline environment;
    (b) global warming will rise sea level;
    (c) global warming will increase the formation of hard shells in ocean; or
    (d) global warming will affect food marine chain.

Section B. (Three questions with unequal marks)

11. (a) Trinitrotoluene (TNT), $C_7H_5N_3O_6$, combines explosively with oxygen to produce CO_2, water, and nitrogen gas. Balance the chemical equation for the reaction and calculate the theoretical oxygen demand for 100 g of TNT.

(6 marks)

(b) Assume that the gas above the soda in a bottle of soft drink is pure CO_2 at a pressure of 2 atm. Calculate the pH of the soft drink. You may realize the solution will be somehow acidic with negligible carbonate.

(12 marks)

12. (a) Some wastewater has a BOD_5 of 150 mg/L at 20 °C. The reaction rate k at that temperature has been determined to be 0.0096/hour. Find the BOD_5 at 15 °C.

(8 marks)

(b) Draw the oxygen sag curve and briefly introduce the curve (< **200 words**, **up to 1 figure** is allowed).

(10 marks)

(c) A sample of water contains various ions including 40.0 mg/L Ca^{2+}, 10.0 mg/L Mg^{2+}, 7.0 mg/L K^+, 110.0 mg/L HCO_3^-, 67.2 mg/L SO_4^{2-}, 11.0 mg/L Cl^-, and certain amount of Na^+. Assuming no other constituents are missing, what would be the concentration of Na^+? What is the total hardness of this water sample in mg/L as $CaCO_3$?

(12 marks)

13. (a) A rectangular, well-mixed aeration lagoon is 60 m long, 5 m wide, and 2 m deep. It receives 400 m^3/d of wastewater with a BOD_5 of 336 mg/L for treatment. The biodegradation rate constants for the wastewater are: endogenous decay constant $k_d = 0.08$ d^{-1}, maximum specific microbial growth rate $\mu_m = 1.10$ d^{-1}, and the half velocity constant $K_s = 76.0$ mg BOD_5/L. What is the efficiency of the lagoon?

(12 marks)

(b) Briefly discuss the global warming as well as its adverse effects. (< **300 words**)

(10 marks)

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